<u>Executive Summary:</u> *Busy Airports, Predicting TSA Traffic at Major U.S. Airports*

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Background and Project Overview:

Airports in the United States serve as critical transportation hubs, handling hundreds of millions of travelers each year. One of the most significant bottlenecks in air travel is the TSA security checkpoint. For TSA directors, accurately forecasting passenger volume is essential for effective staffing and resource allocation. Reliable predictions of daily throughput can also help travelers anticipate longer-than-usual wait times and plan accordingly.

The goal of this project is to develop a predictive model for passenger throughput at TSA security checkpoints. We began by building and validating the model for a single airport, *Chicago O'Hare International Airport*, to establish feasibility and performance. The model is intended to forecast total passenger volume over selected future timescales (daily or weekly totals) with the flexibility to evaluate which forecasting window (days or weeks) yields the most reliable results. Once the single-airport model was shown to be performing well, we extended the framework to other major U.S. airports to test the robustness of the modeling methods.

Stakeholders

The primary stakeholders for this project are TSA leadership and airport operations staff responsible for managing security checkpoint performance, including staffing, lane availability, and day-to-day operations. For these groups, accurately forecasting passenger throughput is essential for matching resources to demand. Better predictions help reduce long wait times, avoid overstaffing, and improve overall operational efficiency through more informed, data-driven decisions. Airlines and travelers also benefit indirectly through fewer delays and more predictable wait times. Overall, this project is valuable because it improves the management of one of the most critical choke points in the air travel system.

Data Collection

The data for this project comes from the publically available TSA throughput dataset (https://www.tsa.gov/foia/readingroom). This data is separated into PDF files containing hourly throughput data for each individual week over the past 4 years. In order to use this data, all of these PDF files need to be downloaded and processed into tabulated CSV data. For the years of data which we use in our analysis (2022-2025), this requires processing about 200 PDF files, each with about 1000 pages of tabulated data. The scripts for reading the PDF files, identifying tabulated data, processing these tables into CSV files, and cleaning up the resulting data are provided in the Github repository.

Modeling Approach

To improve on simple baseline methods, we developed two forecasting approaches designed to capture both long-term patterns and short-term fluctuations in passenger volume. Both approaches were applied to daily and weekly data and evaluated independently.

The first approach (*Harmonic + ARIMA*) identifies recurring seasonal patterns, such as predictable changes throughout the year, and combines them with a statistical model that captures shorter-term trends and variations. This allows us to account for both broad seasonal behavior and day-to-day or week-to-week changes in passenger flow.

The second approach (*STL + ARIMA*) separates the data into three components: overall trend, seasonal patterns, and random variation. We then forecast the underlying trend and variability using a statistical model, while projecting seasonal patterns based on historical behavior. This structured approach helps isolate and better predict different drivers of passenger volume.

For both methods, model settings were automatically optimized to ensure strong performance without manual tuning. We tested and compared the models using multiple rolling evaluation periods to simulate real-world forecasting conditions and identify the most reliable approach.

Finally, we evaluated the selected models on recent unseen data (the last two quarters of 2025) by generating forward-looking forecasts and comparing them to actual outcomes. This allowed us to confirm that the models perform consistently and provide reliable predictions for operational use.

Results

Our analysis found that the Harmonic ARIMA model provided the most accurate daily forecasts for TSA passenger throughput at Chicago O’Hare International Airport. When evaluated on unseen test data (Q3 and Q4 of 2025), the model substantially outperformed both baseline methods and alternative forecasting approaches, where the model achieved an average error of under 10% – accuracy unmatched by any of the other models under consideration.

To assess how well the model generalizes, we applied it to several other major U.S. airports, including JFK, LAX, DFW, DTW, and DEN. The model maintained strong performance at large, high-traffic airports such as JFK, LAX, and DFW, demonstrating its robustness in similar operational environments. While performance was slightly worse at DTW and DEN, it still exceeded that of baseline/alternative models by a meaningful margin.

These results suggest that the model is particularly effective for major airports and could be reliably extended beyond the initial case study, with geographic differences playing a smaller role than airport scale and traffic volume.

Conclusions

This project demonstrates that accurate, data-driven forecasting of TSA passenger throughput is both feasible and operationally valuable. The strong performance of the Harmonic ARIMA model at Chicago O’Hare and other major airports shows its potential to support better staffing decisions, reduce congestion, and improve the overall passenger experience.

Looking ahead, this framework can be extended to a broader set of airports by combining time series methods with additional features such as airport size, geographic data, and regional demand. This would enable more generalizable predictions across different locations. Further improvements may also come from exploring alternative approaches, including models like Prophet or gradient boosting machine learning methods such as XGBoost.